Assessment- 1

For this project, I built a Retrieval-Augmented Generation (RAG)-based Prerequisite and Course Planning Assistant using publicly available academic catalog documents as the knowledge base. The dataset consists of course descriptions, program requirements, and academic policy documents, ensuring coverage of prerequisite chains, program rules, and edge cases such as exceptions and co-requisites. These documents were ingested using a PDF loader and preprocessed through a chunking strategy with a chunk size of 500 and overlap of 50 to balance context preservation and retrieval efficiency.

The system architecture follows a standard RAG pipeline implemented using LangChain. It includes document ingestion, text splitting, embedding generation using a lightweight HuggingFace model (all-MiniLM-L6-v2), and storage in a FAISS vector database for efficient similarity search. A retriever is configured with top-k results to fetch the most relevant chunks for a given query. The retrieved context is then passed to a Large Language Model (Gemini) along with a structured prompt that enforces strict grounding, citation inclusion, and safe abstention when information is missing.

The prompt design plays a critical role in ensuring reliability. It explicitly instructs the model to use only the provided context, include citations for every claim, ask clarifying questions when required inputs are missing, and avoid hallucination by responding with “I don’t have that information” when necessary. This ensures that the assistant adheres to the requirements of correctness and grounded reasoning as specified in the assessment.

For evaluation, a test set of queries can be constructed covering prerequisite validation, multi-step prerequisite chains, program requirement queries, and edge cases where information is not present in the documents. The system’s performance can be measured using metrics such as citation coverage rate, correctness of eligibility decisions, and accuracy of abstention in “not in catalog” scenarios. Initial observations indicate that the system performs well on direct prerequisite queries but may face limitations when handling ambiguous queries or incomplete student profiles, where clarifying questions become essential.

Key challenges encountered include handling large document sizes (which increased processing time), ensuring consistent citation formatting, and managing cases where relevant information is spread across multiple documents. These were addressed by optimizing chunk size, using a faster embedding model, and enforcing strict prompt constraints. Future improvements could include adding caching for faster startup, implementing a multi-agent architecture for better reasoning separation (e.g., retriever, planner, verifier), and enhancing the evaluation pipeline with automated scoring.

Overall, the system demonstrates a practical implementation of a grounded RAG-based assistant capable of answering prerequisite queries and assisting in course planning while maintaining reliability, transparency, and safe behavior.